



## Diesel Generator Set

# mtu 12V4000 DS1750

380V – 11 kV/50 Hz/prime power for stationary emergency/  
NEA (ORDE) optimized/12V4000G14F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

## Product highlights

### Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

### Support

- Global product support offered

### Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

### Power rating

- System ratings: 1590 kVA - 1700 kVA
- Accepts rated load in one step per NFPA 110\*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5\*

### Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

### Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

### Emissions

- NEA (ORDE) optimized

### Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

\* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce  
solution

Application data <sup>1)</sup>

| Engine                         |             |       | Liquid capacity (lubrication)              |     |
|--------------------------------|-------------|-------|--------------------------------------------|-----|
| Manufacturer                   | mtu         |       | Total oil system capacity: l               | 260 |
| Model                          | 12V4000G14F |       | Engine jacket water capacity: l            | 160 |
| Type                           | 4-cycle     |       | Intercooler coolant capacity: l            | 40  |
| Arrangement                    | 12V         |       | Combustion air requirements                |     |
| Displacement: l                | 57.2        |       |                                            |     |
| Bore: mm                       | 170         |       |                                            |     |
| Stroke: mm                     | 210         |       | Combustion air volume: m³/s                | 1.8 |
| Compression ratio              | 16.4        |       | Max. air intake restriction: mbar          | 50  |
| Rated speed: rpm               | 1500        |       | Cooling/radiator system                    |     |
| Engine governor                | ECU 9       |       |                                            |     |
| Max power: kWm                 | 1420        |       |                                            |     |
| Air cleaner                    | dry         |       | Coolant flow rate (HT circuit): m³/hr      | 56  |
|                                |             |       | Coolant flow rate (LT circuit): m³/hr      | 30  |
|                                |             |       | Heat rejection to coolant: kW              | 545 |
|                                |             |       | Heat radiated to charge air cooling: kW    | 260 |
|                                |             |       | Heat radiated to ambient: kW               | 75  |
| Fuel system                    |             |       | Fan power for electr. radiator (40°C): kW  | 55  |
| Maximum fuel lift: m           | 5           |       | Exhaust system                             |     |
| Total fuel flow: l/min         | 16          |       |                                            |     |
|                                |             |       |                                            |     |
| Fuel consumption <sup>2)</sup> | l/hr        | g/kwh | Exhaust gas temp. (after turbocharger): °C | 505 |
| At 100% of power rating:       | 342.2       | 200   | Exhaust gas volume: m³/s                   | 4.9 |
| At 75% of power rating:        | 274.6       | 214   | Maximum allowable back pressure: mbar      | 85  |
| At 50% of power rating:        | 200.2       | 234   | Minimum allowable back pressure: mbar      | 30  |

Standard and optional features

System ratings (kW/kVA)

| Generator model                                                 | Voltage | NEA (ORDE) optimized |      |      |                          |      |      |
|-----------------------------------------------------------------|---------|----------------------|------|------|--------------------------|------|------|
|                                                                 |         | without radiator     |      |      | with mechanical radiator |      |      |
|                                                                 |         | kWel                 | kVA* | AMPS | kWel                     | kVA* | AMPS |
| Leroy Somer LSA52.3 S5<br>(Low voltage<br>Leroy Somer standard) | 380 V   | 1360                 | 1700 | 2583 | 1296                     | 1620 | 2461 |
|                                                                 | 400 V   | 1360                 | 1700 | 2454 | 1296                     | 1620 | 2338 |
|                                                                 | 415 V   | 1360                 | 1700 | 2365 | 1296                     | 1620 | 2254 |
| Marathon 743RSL7090<br>(Low voltage Marathon)                   | 380 V   | 1352                 | 1690 | 2568 | 1304                     | 1630 | 2477 |
|                                                                 | 400 V   | 1344                 | 1680 | 2425 | 1304                     | 1630 | 2353 |
|                                                                 | 415 V   | 1272                 | 1590 | 2212 | 1272                     | 1590 | 2212 |
| Marathon 744RSL7091<br>(Low voltage Marathon<br>oversized)      | 380 V   | 1352                 | 1690 | 2568 | 1304                     | 1630 | 2477 |
|                                                                 | 400 V   | 1344                 | 1680 | 2425 | 1304                     | 1630 | 2353 |
|                                                                 | 415 V   | 1272                 | 1590 | 2212 | 1272                     | 1590 | 2212 |
| Marathon 1020FDH7095<br>(Medium volt. marathon)                 | 11 kV   | 1344                 | 1680 | 88   | 1296                     | 1620 | 85   |
| Leroy Somer LSA53.2 VL6<br>(Medium volt. Leroy Somer)           | 11 kV   | 1352                 | 1690 | 89   | 1296                     | 1620 | 85   |

\* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).  
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.